Sagar Prasad

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Full-stack developer with open-source contributions to data-engineering tooling and a filed patent in palm-biometric presentation-attack detection.

Skills

Frontend: React.js, Next.js, Tailwind CSS, GSAP, Vite
Backend & Data: Node.js, Express.js, FastAPI, REST APIs, PostgreSQL, MongoDB, Supabase
Tools & Testing: Git, GitHub Actions, Docker, Linux, CI/CD, Vercel, Playwright, Pytest, TestSprite

Experience

Freelance Web Developer Remote Ongoing

- Delivered production landing pages and frontend interfaces in React and Next.js for direct clients.

Undergraduate Student Researcher DIT University, Dehradun May 2025 – May 2026

- Researched palm biometrics and feature fusion for presentation-attack detection; true and first inventor on a filed patent.

Technical Coordinator, 38th National Games India Feb 2025

- Led Technical Screening & Recording (TSR) and Game Management System (GMS) operations over 15 days for the sport of Wushu (live results, brackets, and schedules) and managed VIP movement and coordination for the sport of Netball.

Open Source

DataLoom — Ceylon Computer Science Institute (C2SI) github.com/c2siorg/dataloom

- Contributed **14+ merged PRs** across the React frontend and FastAPI backend, including:
- Authored the project's first **Playwright** end-to-end suite, covering all 5 core user flows (upload, transformations, checkpoints, revert, export) and fixing multiple backend defects it surfaced (PR #309).
- Hardened the CSV upload path with **chunked-streaming** validation and consolidated size limits into a single config source, removing a memory-exhaustion vector that buffered full uploads before validating (PR #278).
- Diagnosed and fixed a **checkpoint/replay** bug that silently dropped *all prior* user transformations across successive saves (PR #275).

Patents

A Biometric Presentation Attack Detection System and Method for Detecting Biometric Presentation Attacks
Published Indian Patent **IN202611004117 A1** | Filed 15 Jan 2026, Published 20 Feb 2026 | True and First Inventor

- Multi-modal liveness detection that fuses **pulse oximetry**, **thermal imaging**, and **microscopic skin imaging** in a sequential verification pipeline; grants authentication only when all three modalities confirm a living subject and rejects artificial or spoofed presentations with a spoof alert. Targets palm-based biometric authentication.

Projects

Eterna geteterna.vercel.app

- Local-first **Manifest V3 Chrome extension** for notes, tasks, events, and people across the side panel and new tab; combines Markdown editing, **autosave with cross-tab conflict detection**, fuzzy search, time-zone-aware reminders, and schema-validated ZIP backup/restore. **Stack:** React, TypeScript, WXT, Tailwind CSS, TipTap, Dexie/IndexedDB, MiniSearch, Valibot, fflate.

postheap postheap.vercel.app

- Published Chrome extension (MV3) + dashboard that backs up Instagram saved posts from the user's own saved-feed API responses, not DOM scraping; token-authed batch ingest, caption search, keyword auto-sort collections. **Stack:** Next.js 14, TypeScript, Supabase (Postgres RLS, Auth, Storage), Chrome MV3.

macfolio bysagar.vercel.app

- Personal portfolio reimagined as a macOS desktop with a dock, draggable windows, and built-in "apps" (Finder, Terminal, Resume PDF viewer); custom Zustand-backed window manager handles open/close, focus, and z-index ordering. **Stack:** React, Vite, Zustand, GSAP Draggable, Tailwind CSS, react-pdf.

Education

DIT University Dehradun, India
Bachelor of Computer Applications 2024 – 2027

Achievements and Honors

- **1st Place**, Idea & Prototype Competition 2026 (CodeGenX), DIT University.
- **Finalist**, Watch the Code 2026 (National-Level Hackathon), Graphic Era Hill University, Haldwani.
- **Responsible Disclosure** reported to Snapchat Inc. and the Delhi Government Education Department (edudel.nic.in)